

Code Audit Notes

This repository intentionally preserves the original MATLAB implementation. The audit below separates execution requirements and obvious issues from changes that would alter the original methodology.

Confirmed source files

The supplied project contains the following MATLAB source files:

- `HNN.m`
- `EPSO.m`
- `UseSVR.m`
- `UseNN.m`
- `UseEPSO.m`
- `irrel_filter.m`
- `redund_filter.m`
- `Output_Label.m`
- `OF.m`

`Contingency_Table.m` was referenced by the supplied source but was not included in the supplied archive. The repository includes a reconstructed version based on its call signature and the metrics documented in the FYP report.

External dependency: LIBSVM

`UseSVR.m` calls `svmtrain` and `svmpredict`. These are not MATLAB base-language functions in this project; the original code expects the LIBSVM MATLAB interface.

The original `HNN.m` also contains a machine-specific `addpath` pointing to the original developer's computer. That path cannot be reused on another computer and is therefore documented as a setup requirement rather than treated as a portable dependency.

Missing random-order files

The original `HNN.m` loads `randomorder_california`. The supplied project archive does not contain that file. The Chile equivalent is also not supplied. No replacement random-order file has been invented because doing so could change the original experiment.

For this reason, the repository does not claim that the original experiment has been reproduced yet.

redund_filter.m execution issue

The supplied implementation can reach `featSet(:, drop_ind)=-1` when `RC(k) <= 0.2`, even though `drop_ind` is only assigned inside the `RC(k)>0.2` condition. This is an execution bug in the supplied source.

It has **not** been silently changed in the preserved source, because fixing it can change the selected feature set and therefore the experiment's results. It should be addressed only in a separately documented reproducibility/fix version after the original source is preserved.

SVR scaling

`UseSVR.m` standardizes the training columns and test columns separately. A modern machine-learning implementation would normally calculate scaling parameters on the training set and apply those same parameters to the test set. This repository intentionally leaves the original behavior unchanged because changing it could change the original results.

Neural-network workflow

`UseNN.m` uses the older shallow neural-network API (`train`, `sim`, and a network created with `newff`). It also uses `trainlm`, `trainbfg`, and `trainbr` in successive stages. These choices are preserved rather than replaced with newer APIs.

EPSO

`EPSO.m` uses a population of 400 particles and 20 generations. Its objective function is `OF.m`, which returns MCC from `Contingency_Table.m`. These parameters and the optimization logic are preserved.

Reproduction status

The code has been statically audited, but it has not yet been executed in MATLAB in this environment. The next reproduction step therefore requires a compatible MATLAB installation plus the original LIBSVM MATLAB interface and, if exact historical results are required, the missing original random-order files.